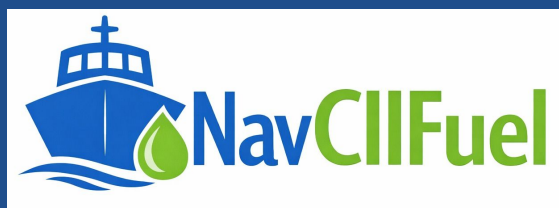


MARITIME COMPLIANCE TOOLS

USER GUIDE

C_F Maritime Calculator — Web Version



CF · Bio fuel blend and GHG

Prepared for: Vessel Compliance & Reporting Teams

Document version: 1.0

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1 Introduction

1.1 Purpose of this guide

This guide explains how to use the CF 905 Biofuel C^f Calculator, a browser-based tool for calculating the carbon factor (C^f) of a biofuel/fossil fuel blend and the resulting CO₂ emissions for a given fuel quantity. It walks through every input field, the auto-calculated fraction breakdown, the results panel, and the toolbar's export functions, using figures based directly on the tool's own layout.

The guide is written for ship operators, bunkering officers, and compliance teams who need to enter blend data, run a C^f / CO₂ calculation, and produce a shareable record.

1.2 What the tool does

- Calculates the carbon factor (C^f) of the biofuel fraction from its lower calorific value (LCV) and emission intensity.
- Calculates the weight (% mass and kg) and energy content of the biofuel and fossil fuel fractions from a single fuel quantity entry.
- Calculates the blended C^f value (energy-weighted average of the biofuel and fossil fuel C^f values) and the total CO₂ emissions for the fuel quantity loaded.
- Supports a single blend entered manually, or multiple blends imported from an Excel template, with per-blend results shown in a table.
- Saves/reloads a full case as an XML file, and produces a print-ready PDF/Word report.

Before you start

- The tool runs entirely in your browser — no data is uploaded or sent anywhere.
- Have the Bunker Delivery Note (BDN) for the biofuel and fossil fuel fractions, and the biofuel's certificate of quality (density, LCV or energy content, and emission intensity), ready before you start.
- Use a current version of Chrome, Edge, or Firefox for the best experience.

2 Interface Overview

The calculator is organised into a fixed toolbar and three numbered sections: A — Inputs, B — Biofuel fraction & Fossil fuel fraction, and C — Results. Each section can be collapsed using the arrow (Λ) in its header.

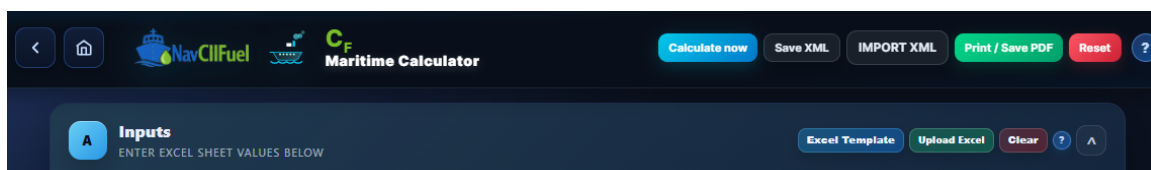


Figure 1 — Toolbar: Calculate now, Save XML, Import XML, Print / Save PDF, Reset.

The toolbar buttons are available at all times and are described in the table below.

Button	Function
Calculate now	Runs the C^f / CO_2 calculation using the current inputs and refreshes Sections B and C.
Save XML	Exports all current inputs (fuel quantity, biofuel fraction data, fossil fuel grade) as an .xml case file for later reuse.
Import XML	Loads a previously saved .xml case file and repopulates all fields automatically.
Print / Save PDF	Opens a formatted, print-ready report and the browser print dialog (choose “Save as PDF” as the destination).
Reset	Clears all entered values in Sections A–C and returns the calculator to a blank state.
? (Page Help)	Opens the built-in on-screen help guide describing every field and formula in Sections A–C.

3 Quick Start Workflow

Follow these five steps for a typical single-blend calculation:

1. Open the calculator HTML file in your browser and confirm the page loads with Sections A and B visible.
2. In Section A, enter the total fuel quantity loaded, then the biofuel fraction quantity (MT, or m³ with density), its total energy and emission intensity.
3. Select the fossil fuel grade from the drop-down in the Fossil fuel fraction subgroup.
4. Click Calculate now and review the auto-calculated fraction breakdown in Section B and the C^f / CO_2 results in Section C.
5. Use Save XML to archive the case and Print / Save PDF to generate the record for distribution.

4 Section A — Inputs

Section A captures the fuel quantity loaded and the details of the biofuel and fossil fuel fractions that make up the blend. It is organised into three subgroup cards: Fuel quantity loaded, Biofuel fraction, and Fossil fuel fraction.

The screenshot shows the 'Section A Inputs' form. At the top, there's a header with 'A Inputs' and a sub-header 'ENTER EXCEL SHEET VALUES BELOW'. To the right are buttons for 'Excel Template', 'Upload Excel', 'Clear', a help icon, and a refresh icon. The form is divided into three main sections:

- Fuel quantity loaded:** A single input field for 'Total fuel quantity loaded' with a unit selector set to 'MT'.
- Biofuel fraction:** This section contains four input fields:
 - 'Quantity of biofuel fraction ^' with a unit selector set to 'MT'.
 - 'Quantity of biofuel fraction ^' with a unit selector set to 'm³'.
 - 'Density of biofuel fraction ^' with a unit selector set to 'kg/m³'.
 - 'Total Energy from biofuel fraction ^' with a unit selector set to 'MJ'.
 - 'Emission intensity of biofuel fraction ^' with a unit selector set to 'g CO₂e/MJ'.
- Fossil fuel fraction:** A single dropdown menu for 'Fuel grade of fossil fraction (select from drop down)*'.

Figure 2 — Section A with sample blend data entered.

Field	Description
Total fuel quantity loaded (MT)	Total mass of the fuel blend loaded, in metric tonnes. Used as the basis for the CO ₂ emissions calculation.
Quantity of biofuel fraction (MT)	Mass of the biofuel fraction, in metric tonnes. Enter this OR the m³ + density pair below — not both.
Quantity of biofuel fraction (m³)	Volume of the biofuel fraction, in cubic metres. Leave blank if the MT value above is entered directly.
Density of biofuel fraction (kg/m³)	Density of the biofuel fraction. Required only when the biofuel quantity is entered as m³ rather than MT.
Total Energy from biofuel fraction (MJ)	Total energy content of the biofuel fraction, in megajoules, from the biofuel supplier's documentation.
Emission intensity of biofuel fraction (g CO₂e/MJ)	Well-to-wake or tank-to-wake emission intensity of the biofuel, per its certificate of quality.
Fuel grade of fossil fraction	Selects the fossil fuel grade (Distillate fuel, Residual fuels RMA–RMD, or Residual fuels RME–RMK). Determines the fossil C ^f and LCV values used

Field	Description
	automatically.
Tips • Enter the biofuel quantity either in MT, or as m³ with a density value — the tool needs one complete pair to calculate the biofuel weight in kg. • If the biofuel's emission intensity exceeds 33 g CO₂e/MJ, the tool automatically applies the distillate fuel C^f value (3.206) to the biofuel fraction instead of the calculated value. • Re-run Calculate now after changing any field so Sections B and C reflect the update.	

4.1 Importing multiple blends from Excel

Use the buttons in the top-right of the Section A header to work with more than one blend at once:

Control	Description
Excel Template	Downloads a blank .xlsx file with the correct column headers for offline entry of multiple biofuel blends.
Upload Excel	Imports a completed .xlsx, .xls, .csv or .html file and displays every blend row in an editable table beneath Section A. Rows can be added or removed individually.
Clear	Removes all uploaded blend data and restores the manual single-blend input form.

5 Section B — Biofuel Fraction & Fossil Fuel Fraction

Section B shows the automatically calculated weight and energy breakdown of the blend, based on the values entered in Section A. It requires no direct input.

B Biofuel fraction & Fossil fuel fraction AUTO-CALCULATED BREAKDOWN		
DESCRIPTION	UNIT	VALUE
Biofuel fraction		
Weight	% mass	10.000000
Weight	kg	50000.000
Calculated lower Calorific Value (LCV) of biofuel	MJ/kg	38.000
Cf value	g-CO ₂ /g fuel	0.760
Energy content of biofuel fraction	MJ	1900000.000
Fossil fuel fraction		
Weight	% mass	90.000000
Weight	kg	450000.000
Cf value	g-CO ₂ /g fuel	3.206
Energy content of fossil fuel fraction	MJ	19215000.000

Figure 3 — Auto-calculated fraction breakdown for the sample blend.

Result	How it is calculated
Weight (% mass) — biofuel	Biofuel kg ÷ total fuel kg × 100.
Weight (kg) — biofuel	Biofuel MT × 1000, or biofuel m ³ × density when MT is not entered directly.
Calculated lower Calorific Value (LCV) of biofuel	Total energy from biofuel fraction (MJ) ÷ biofuel weight (kg).
C^f value — biofuel (g-CO₂/g fuel)	(LCV ÷ 1000) × emission intensity — unless emission intensity exceeds 33 g CO ₂ e/MJ, in which case the distillate value 3.206 is used.
Energy content of biofuel fraction (MJ)	The total energy from the biofuel fraction as entered in Section A.
Weight (% mass / kg) — fossil fuel	The remainder of the total fuel quantity after the biofuel weight is subtracted.
C^f value — fossil fuel (g-CO₂/g fuel)	Fixed value for the selected fossil fuel grade (Distillate: 3.206; RMA–RMD: 3.151; RME–RMK: 3.114).
Energy content of fossil fuel fraction (MJ)	Fossil fuel weight (kg) × LCV of the selected fossil fuel grade.

6 Section C — Results

Section C appears after Calculate now is clicked. It shows the final blended carbon factor and the total CO₂ emissions for the fuel quantity loaded.

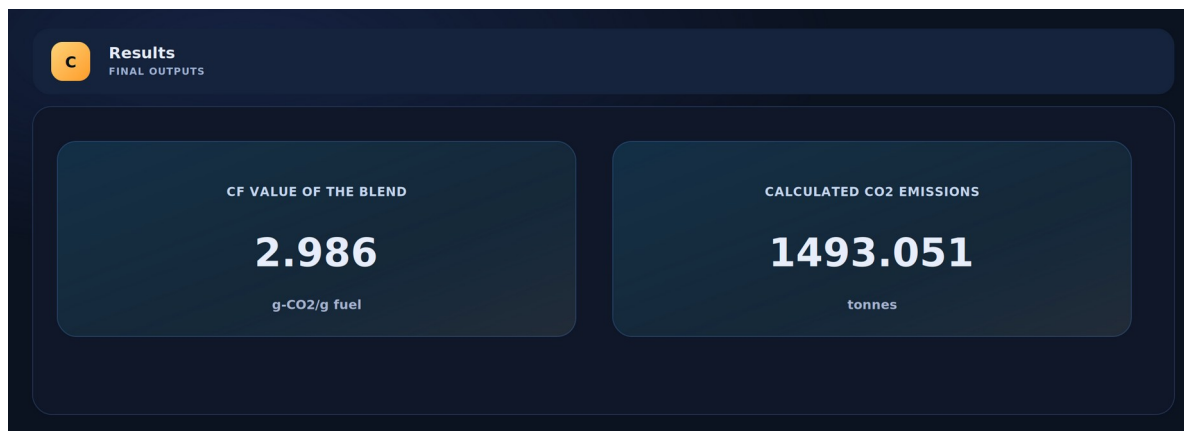


Figure 4 — Calculated Cf value and CO₂ emissions for the sample blend.

Result	Meaning
C^f value of the blend (g-CO₂/g fuel)	Energy-weighted average of the biofuel and fossil fuel C ^f values: (biofuel energy share × biofuel C ^f) + (fossil energy share × fossil C ^f).
Calculated CO₂ emissions (tonnes)	Blend C ^f value × total fuel quantity loaded (MT).

When multiple blends have been imported via Excel, Section C instead shows a Multiple biofuel blend results table listing the blend name, bio energy (MJ), fuel grade, blend C^f and CO₂ tonnes for every blend, calculated separately.

7 Exporting and Record-Keeping

- Save XML after completing each case, to preserve every input value for future reuse or audit.
- Import XML to reload a saved case rather than re-entering data.
- Use Print / Save PDF once the results have been checked, choosing “Save as PDF” in the browser print dialog.
- Keep an editable Word copy of the report alongside the PDF for internal review and version tracking.
- Name saved files consistently, e.g. VesselName_BDNRef_CF.xml / .pdf, for easy retrieval.

8 Best-Practice Checklist

- Enter the biofuel quantity as either MT or (m³ + density) — never leave both pairs blank.
- Double-check the total energy and emission intensity values against the biofuel supplier's certificate of quality.
- Confirm the correct fossil fuel grade is selected, as this determines the fossil C^f and LCV values used.

- After every calculation, review the fraction breakdown in Section B before relying on the Section C results.
- Store the XML input file together with the generated PDF/Word report for full traceability.

9 Troubleshooting

Symptom	Likely cause / fix
Results in Section C show “—”	Calculate now has not been run yet, or a required field in Section A is empty — fill all required fields and click Calculate now.
Biofuel weight (kg) shows “—”	Neither a complete MT value nor a complete m ³ + density pair has been entered for the biofuel fraction.
C ^f value of biofuel equals 3.206 unexpectedly	This is expected behaviour when the emission intensity entered exceeds 33 g CO ₂ e/MJ; the tool substitutes the distillate C ^f value.
Uploaded Excel file did not populate all rows	Check that the file was created from the downloaded Excel Template and that all required columns are filled in.
Print / Save PDF opens a blank or partial report	Allow the browser a few seconds to render before printing, and ensure pop-ups are not blocked for this file.

Appendix — Full Interface Reference

The figures below show the calculator's toolbar and each numbered section with sample data entered and a calculation already run, for reference alongside the section-by-section descriptions in this guide.

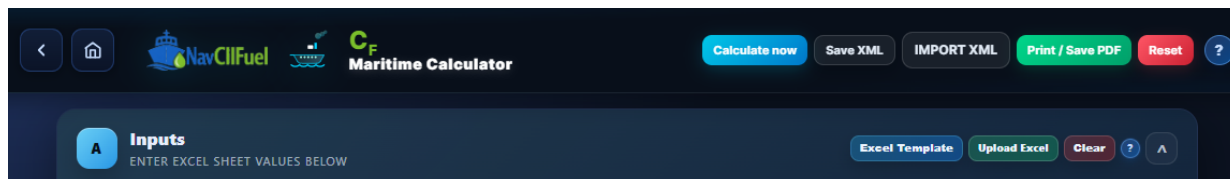


Figure 1 (repeated) — Toolbar.

A Inputs ENTER EXCEL SHEET VALUES BELOW Excel Template Upload Excel Clear ? A							
EXCEL UPLOAD ACTIVE 13 blends Add row Uploaded blend data shown in Part A Values imported from Excel are displayed below. Values are editable and shown with two decimal places. Use Add row to include another blend or Remove to delete an uploaded blend row.							
Blend	Total fuel mt	Biofuel mt	Biofuel m ³	Density kg/m ³	Bio energy mj	Bio ei	Fuel grade
Blend 1	1200.00	200.00	0.00	0.00	9768148.00	9.20	Distillate fuel
Blend 2	1250.00	250.00	0.00	0.00	9768148.00	9.30	Distillate fuel
Blend 3	1300.00	300.00	0.00	0.00	9768148.00	9.40	Distillate fuel
Blend 4	1350.00	350.00	0.00	0.00	9768148.00	9.50	Distillate fuel
Blend 5	1400.00	400.00	0.00	0.00	9768148.00	9.60	Distillate fuel
Blend 6	1450.00	450.00	0.00	0.00	9768148.00	9.70	Distillate fuel
Blend 7	1500.00	500.00	0.00	0.00	9768148.00	9.80	Distillate fuel
Blend 8	1550.00	550.00	0.00	0.00	9768148.00	9.90	Distillate fuel
Blend 9	1600.00	600.00	0.00	0.00	9768148.00	10.00	Distillate fuel
Blend 10	1650.00	650.00	0.00	0.00	9768148.00	10.10	Distillate fuel
Blend 11	1700.00	700.00	0.00	0.00	9768148.00	10.20	Distillate fuel
Blend 12	1750.00	750.00	0.00	0.00	9768148.00	10.30	Distillate fuel
Blend 13	1800.00	800.00	0.00	0.00	9768148.00	10.40	Distillate fuel

Figure 2 (repeated) — Section A: Inputs.

B Biofuel fraction & Fossil fuel fraction AUTO-CALCULATED BREAKDOWN		
DESCRIPTION	UNIT	VALUE
Biofuel fraction		
Weight	% mass	10.000000
Weight	kg	50000.000
Calculated lower Calorific Value (LCV) of biofuel	MJ/kg	38.000
Cf value	g-CO2/g fuel	0.760
Energy content of biofuel fraction	MJ	1900000.000
Fossil fuel fraction		
Weight	% mass	90.000000
Weight	kg	450000.000
Cf value	g-CO2/g fuel	3.206
Energy content of fossil fuel fraction	MJ	19215000.000

Figure 3 (repeated) — Section B: Biofuel fraction & Fossil fuel fraction.

C Results FINAL OUTPUTS				
Multiple biofuel blend results Each uploaded blend is calculated and displayed separately.				
BLEND	BIO ENERGY MJ	FUEL GRADE	BLEND C _f	CO ₂ TONNES
Blend 1	9,768,148.000	Distillate fuel	2.693	3,231.340
Blend 2	9,768,148.000	Distillate fuel	2.677	3,345.975
Blend 3	9,768,148.000	Distillate fuel	2.666	3,465.945
Blend 4	9,768,148.000	Distillate fuel	2.658	3,588.962
Blend 5	9,768,148.000	Distillate fuel	2.653	3,713.885
Blend 6	9,768,148.000	Distillate fuel	2.648	3,840.078
Blend 7	9,768,148.000	Distillate fuel	2.645	3,967.160
Blend 8	9,768,148.000	Distillate fuel	2.642	4,094.889
Blend 9	9,768,148.000	Distillate fuel	2.639	4,223.102
Blend 10	9,768,148.000	Distillate fuel	2.637	4,351.689
Blend 11	9,768,148.000	Distillate fuel	2.636	4,480.569
Blend 12	9,768,148.000	Distillate fuel	2.634	4,609.683
Blend 13	9,768,148.000	Distillate fuel	2.633	4,738.988

Figure 4 (repeated) — Section C: Results.

End of user guide.